DATA 202 — Forum 4

Ch. 4 — "How Numbers Get Their Clout"

Group members (all three names):

1. _____ 2. _____ 3. _____

Week 10 — Friday class

Stone's argument: numbers carry an authority that has little to do with how accurate they are. Stone sensed that Boston's traffic had gotten worse, but she believed it only when the *Boston Globe* printed numbers. "Numbers acquire their power the same way the gods acquire theirs—humans invest them with virtues they want their rulers to have. […] Or, as many people do now, call it objectivity." Before standard measures, millers and lords heaped or leveled their measuring containers to their own advantage, and the metric system's platinum meter stick promised a measure that "couldn't play favorites." Yet no measuring system enforces itself, and every count rests on human choices: James Madison argued that slaves should count as three-fifths of a person, and called it "their true character." Today, risk algorithms in criminal justice and child welfare are sold as "high-tech platinum sticks," but their data record human decisions: stops, arrests, convictions, hotline calls. "The subjectivity that algorithms supposedly avoid resides deep in their data sets, only now it's much harder to see." The COMPAS algorithm used 137 pieces of information about each defendant, yet a simple model using just 2 of them predicted as well. Teacher ratings that "control for" poverty show that "scientific 'control for' and human 'control over' are two different animals." Stone's verdict: "Numbers aren't tyrants; they're tyrants' weapons," and "Social issues aren't math problems."

This week we turned houses, abalone and cars into lines: coefficients that hold the other columns fixed; MAE, RMSE and R^2; and penalties that decide which columns a model keeps. This forum asks where those numbers get their clout.

Discuss and write a response to each question below (~25 min).

1. An assessor's model says your house is worth $239,000, though it is typically off by about $38,000. A realtor who walked through your house says $200,000. Which number would most people believe, and why? Should they?

2. Monday's model priced each house "holding the neighborhood fixed." Stone warns that "control for" is not "control over." What does treating the neighborhood as a fixed fact hide about how neighborhoods, and their prices, came to be?

3. COMPAS used 137 facts about each defendant, yet a model with just 2 (age and number of past convictions) predicted just as well. Lasso trims columns in the same way. Should a model with fewer columns earn more of our trust, or less? What can the 2 remaining columns still hide?

Submission. Photograph or scan your annotated worksheet and upload it to Moodle by **Wednesday of the following week (Week 11)**. See Forum Instructions for the full process.